EXPERIMENT NO 03

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NAME: YUGITHA B

REG NO: 192424102

SLOT: A

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SUBJECT: DEEP LEARNING

SEMESTER: MS2-SEPTEMBER-2026

Probability Distributions and Bayes' Theorem

SCENARIO

To generate synthetic hospital patient waiting-time data using Python, calculate probabilities for specified intervals using the normal distribution, and visualize the distribution using a histogram and probability density curve.

AIM

To calculate the covariance matrix of patients' height, weight, and blood pressure using Python and NumPy, determine the eigenvalues and eigenvectors, and visualize the relationships among the features.

ALGORITHM

1. Import the required libraries.
2. Generate synthetic waiting-time samples using a normal distribution.
3. Calculate the mean and standard deviation.
4. Calculate probabilities for specified waiting-time intervals.
5. Plot the histogram and normal distribution curve.
6. Display the results and interpret the probabilities.

SOURCECODE

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Generate synthetic waiting-time data
np.random.seed(42)
waiting_times = np.random.normal(loc=30, scale=8, size=1000)
```

```
waiting_times = np.clip(waiting_times, 0, None)

# Calculate mean and standard deviation
mean = np.mean(waiting_times)
std = np.std(waiting_times, ddof=1)
print("Mean Waiting Time:", round(mean, 2), "minutes")
print("Standard Deviation:", round(std, 2), "minutes")

# Calculate probabilities using normal distribution
intervals = [(0, 20), (20, 40), (40, 50)]
print("\nPROBABILITY RESULTS")
for low, high in intervals:
    probability = norm.cdf(high, mean, std) - norm.cdf(low, mean, std)
    print(f"P({low} to {high} minutes) = {probability:.4f} "
          f"({probability * 100:.2f}%)")

# Plot histogram and normal distribution curve
x = np.linspace(0, 60, 500)

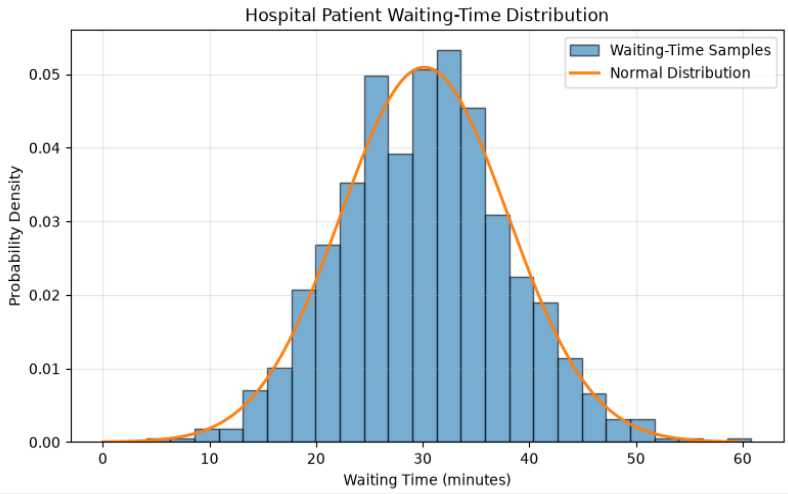
plt.figure(figsize=(9, 5))
plt.hist(waiting_times, bins=25, density=True,
         alpha=0.6, edgecolor='black', label='Waiting-Time Samples')
plt.plot(x, norm.pdf(x, mean, std),
         linewidth=2, label='Normal Distribution')
plt.xlabel("Waiting Time (minutes)")
plt.ylabel("Probability Density")
plt.title("Hospital Patient Waiting-Time Distribution")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```

```
# Summary statistics
print("\nSUMMARY STATISTICS")
print("Minimum:", round(np.min(waiting_times), 2), "minutes")
print("Maximum:", round(np.max(waiting_times), 2), "minutes")
print("Mean:", round(mean, 2), "minutes")
print("Median:", round(np.median(waiting_times), 2), "minutes")
```

OUTPUT

```
Mean Waiting Time: 30.15 minutes
Standard Deviation: 7.83 minutes

PROBABILITY RESULTS
P(0 to 20 minutes) = 0.0974 (9.74%)
P(20 to 40 minutes) = 0.7981 (79.81%)
P(40 to 50 minutes) = 0.0988 (9.88%)
```




```
SUMMARY STATISTICS
Minimum: 4.07 minutes
Maximum: 60.82 minutes
Mean: 30.15 minutes
Median: 30.2 minutes
```

RESULT

Thus, synthetic hospital waiting-time data was generated, and probabilities for specified intervals were calculated using the normal distribution. The histogram and density curve visualize the distribution of patient waiting times and help estimate the likelihood of patients waiting within particular time intervals.